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The Impact of Type of Examples on Originality: Explaining Fixation and Stimulation Effects

ABSTRACT

There are obstacles to creativity: one of them is called fixation effect, the fact that some knowledge about existing or obvious solutions is spontaneously activated and constrains the generation of new solutions. Converging evidence in cognitive psychology has indicated that the ability to generate original ideas can be limited by recently activated knowledge, such as examples of solutions. On the other hand, neuroimaging studies have recently demonstrated that exposure to examples could, on the contrary, have a stimulating effect on originality. To make sense of what seems to be contradictory studies, we hypothesized that two types of examples could have opposite effects on originality: (1) restrictive examples—within the fixation effect—could lead to a reduction in the originality of the solutions, whereas (2) expansive examples—outside the fixation effect—could provoke solutions of higher originality. Results from a total of 160 participants confirmed that the solutions proposed by the group exposed to restrictive examples were less original than those given by the group exposed to expansive examples.

Keywords: creative behavior, fixation effect, design fixation.

Many studies have highlighted how people are likely to face difficulties during creative situations. Purcell and Gero (1996) studied cognitive bias in design situations and showed how fixation and conformity effects can occur in design processes, when individuals have to design new objects (products or services) to fulfill specific functions. This study stressed how individual designers can be trapped by an existing (or an obvious) solution, which constrains the generation of alternative solutions. This is in line with Jansson and Smith (1991) who have shown that the first

explored solution in a design task influences heavily the exploration of new solutions. This phenomenon is characterized as fixation effect, and design literature now proposes practical methodologies to overcome those fixation effects in design contexts (Hatchuel, Le Masson, & Weil, 2011).

In parallel of this research in the design field, trends in cognitive psychology have clarified some obstacles that most people are likely to face during creative situations (Finke, Ward, & Smith, 1992; Ward, Patterson, & Sifonis, 2004). More specifically, numerous reports have indicated that recently activated knowledge can constrain the ability to generate original ideas (Abraham & Windmann, 2007; Abraham, Windmann, Siefen, Daum, & Güntürkün, 2006). For example, creative problem solving can be inefficient when the solution requires subjects to generate an atypical object function when the object's typical function has been primed (Defeyter & German, 2003; Duncker, 1945). This effect is called "functional fixedness". For instance, in the "candle problem" (Adamson, 1952; German & Barrett, 2005), subjects are provided with a candle, a book of matches and a box of tacks, and are asked to affix the candle to a vertical screen in such a way that it will burn without dripping wax onto the table below. The solution to this problem requires subjects to use the box of tacks in an unfamiliar way (i.e., as a platform). Adults fail to spontaneously find this solution when the box is presented as being full of tacks. In other words, when subjects are primed with the typical function of the box, it blocks their ability to use the box in an atypical manner. Traditional interpretations of functional fixedness suggest that accumulated knowledge about the typical properties of the object is automatically activated and blocks the alternative uses of the object. Here, people fail to see new ways of using objects and are not able to find an original solution to problems because they remain bound to a previously activated prescription for use of the object. In the same vein, converging evidence reports a fixation bias when subjects are asked to generate new ideas after being exposed to examples of a solution. Smith, Ward, and Schumacher (1993) studied the constraining effects of using examples on generating creative ideas. Participants were asked to imagine and draw animals that lived on another planet (animals that were very different from those found on Earth). Prior to drawing the animal, participants were exposed to examples that had fundamental elements in common (e.g., eyes or antennas). Two independent experimenters noted whether the subjects' drawings contained any of the previously viewed elements. The results showed that participants tended to incorporate these elements into their own drawings, despite an explicit warning to avoid replicating features given in the examples (see also Jansson & Smith, 1991; Landau & Leynes, 2004; Landau, Thomas, Thelen, & Chang, 2002).

Nevertheless, exposure to examples does not systematically lead to fixation. In the domain of brainstorming, it has been shown that exposure to others' ideas can stimulate associations and lead to the generation of additional ideas. Dugosh and Paulus (2005) reported that participants generated more unique solutions when a list of eight ideas was given prior to a brainstorming session. In addition, using a creative idea generation task in which participants were requested to generate alternative uses of conventional everyday objects such as an umbrella, Fink et al. (2010) reported

that creative performance increased after exposure to examples of solutions. Together, these results suggested that exposure to examples of a solution allowed participants to produce more original ideas.

It should be noted that these studies did not provide clues for understanding why examples lead to fixation effects in some situations, whereas they stimulate creativity in others. It is possible that several types of examples could have different impacts on creativity. To our knowledge, the impact of different types of examples on creativity has never been studied. Although the discrepancies in the results of the aforementioned studies underscore the need for further research in this field, little is known about the potential impact of the nature of examples on the realm of creativity. As a matter of fact, examining the precise influence of different types of examples on idea generation may provide experimental arguments enabling the reconciliation of studies showing apparent opposite results (either stimulating or constraining effects of examples). The design literature on fixation effect has proposed that the activation of spontaneously used knowledge leads to fixation effects. Consequently, it is possible that external cues reinforce the activation of this knowledge and increase the fixation effect. In contrast, examples of less spontaneously accessible solutions activate knowledge and reduce the fixation effect, therefore stimulating originality. We then hypothesize that (1) example within the fixation effect (i.e., solutions generated using spontaneously activated knowledge) constrains creativity and (2) example outside the fixation effect (i.e., solutions generated using knowledge that was not spontaneously activated) stimulates creativity. We label example within the fixation effect as “restrictive example,” and we label example outside fixation as “expansive example.” We chose to use the terms restrictive and expansive in the meaning used in design literature. The term “restrictive” refers to a limitation to the space of possibilities without changing the definition or attributes of the object to design. The term “expansive” highlights a change in the identity of the object by adding unexpected attributes to the initial concept (Hatchuel, 2001). Bridging more closely to design literature, theoretical work has already proposed insights on fixation effect. Specifically, according to C-K theory, people tend to generate ideas that are most accessible in memory. This activation of common knowledge could lead to fixation effects (Hatchuel et al., 2011). Consequently, it is possible that external cues reinforcing the activation of this common knowledge increase the fixation effect. In contrast, examples of solutions that activate knowledge that are less spontaneously accessible should reduce the fixation effect and therefore stimulate originality.

To test these two hypotheses, we used a creative task in which participants had to design a way to drop a hen’s egg from a height of 10 m, so that it would not break. This task was employed because it required minimal engineering expertise and allowed many possible solutions. The investigation was divided into two studies: identifying fixation effect to build both restrictive and expansive example (study 1), and examining the potential impacts on creativity of both types of examples (study 2). The first study aimed at validating both a fixation path based on spontaneously activated knowledge and an expansive path based on activating knowledge that is

less accessible. Thus, a group of participants was asked to give solutions to the egg task. We then used this natural distribution of solutions to ensure that expansive solutions required an expansion of knowledge (i.e., the activation of a set of knowledge that is not spontaneously activated when designing a solution to the egg task). The results of the first study allowed the identification of restrictive and expansive examples. We then examined the specific impact of these examples on originality in a second study, in which two groups of participants were asked to solve the egg task after being given either of those examples.

STUDY 1: IDENTIFYING FIXATION EFFECT METHOD

To identify fixation effect in this creative task, we had to acknowledge the set of solutions generated using spontaneously activated knowledge, and we had also to acknowledge the alternative set of solutions. To do so, we relied on the most recent development in design theory to model the fixation phenomenon, namely C-K theory. In empirical contexts, C-K theory has shown its potential not only to describe existing paths of creative solutions but also to model the generation of ideas outside existing paradigms (Agogu  , Le Masson, & Robinson, 2012).

Therefore, in the field of design science, the development of Concept-Knowledge design theory (C-K theory) models creative reasoning (Hatchuel & Weil, 2009; Kazakçı & Tsoukias, 2005; Le Masson, Weil, & Hatchuel, 2010; Reich, Hatchuel, Shai, & Subrahmanian, 2012) and offers a new and rich model of fixation effects (Hatchuel et al., 2011). The C-K design theory allows modeling the creative process as the interrelated expansion of two spaces (Hatchuel & Weil, 2009; see Figure 1). One space, defined as the Concept Space (C-Space), is tree-structured and describes

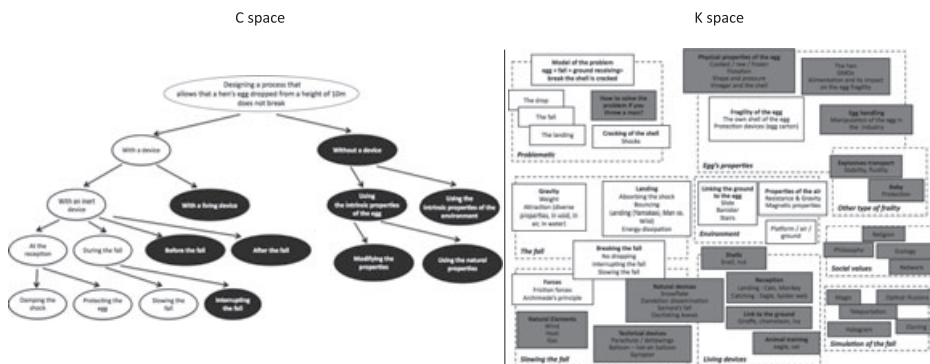


FIGURE 1. The knowledge in the design task classically activated is represented in white, whereas the gray boxes represent the expansion of knowledge in the K-space. In the C-space, the white paths are the restrictive partitions proposed to solve the design task and the gray ones are the expansive partitions.

the progressive and stepwise generation of alternatives. The other space, defined as the Knowledge Space (K-Space), is formed by the network of memorized and activated knowledge that is used for the generative process of concepts in the Concept Space. C-K theory then sets the framework for a design process based on refining and expanding an initial concept by adding attributes stemming from the K-space. It offers then a way to characterize different paths of solutions and the pockets of knowledge associated with the different sets of solutions.

To use C-K theory, we first gathered the knowledge and the solutions usually proposed for the egg task (such as using shells, mattresses, or parachutes). Our hypothesis was that this knowledge base is the one that is classically activated while solving the task. We then expanded into the C-space by describing the partitions that had been made to achieve the design of devices that change either the receptacle or the fall, that is, those solutions to the problem of dropping an egg from a 10-meter height without breaking it that use a device that is inert (i.e., not living). The expansions (1) without using a device and (2) using a living device lead us to expand into the K-space to look for new knowledge (e.g., about the egg's properties, about living devices). It allowed us to think of new expansions in the C-space (i.e., using the natural properties of the egg or modifying them). The use of C-K theory allowed us to map the possible solutions axis, as shown in the Figure 1.

According to C-K theory, this model helped us to think of possible paths to a solution that did not come spontaneously (e.g., training an eagle to catch the egg, using the natural robustness of the longitudinal axis of an egg, freezing the egg). In the first experiment, we examined whether C-K reasoning indeed helped to identify fixation effect.

PARTICIPANTS

Twenty-eight undergraduate students from Paris Descartes University were recruited for this study. Participants were between 17 and 22 years old (mean age: 18.8 years).

PROCEDURE

Each participant was given 10 min to generate as many original solutions as possible to the following problem: "Ensure that a hen's egg dropped from a height of 10 m does not break."

RESULTS

As illustrated in Table 1, analysis of the responses given by the participants revealed a strong fixation effect: 81% of the responses consisted of using an inert device to dampen the shock (33%), protect the egg (26%), or slow the fall (22%) (χ^2 significant at $p < .005$). One good example of a solution following the fixation path based on spontaneously activated knowledge was to slow the fall with a parachute. On the contrary, solutions belonging to expansive paths were less often proposed by the participants (19%, χ^2 significant at $p < .005$). Interestingly, fewer than 10% of the responses consisted of using a living device (4%) or of modifying the

TABLE 1. Natural Distribution of Solutions to the Egg Task

Categories	Example of solution	Occurrence (%)
Damping the shock	Using a mattress	33
Protecting the egg	Using cotton around the egg to protect it	26
Slowing the fall	Hanging the egg to a parachute	21
Interrupting the fall	Using a net a few centimeters below the launch	7
Acting before the fall	Letting go of the egg at a height of 11 m	2
Acting after the fall	Replacing the broken egg by a sound one	0
Using a living device	Training an eagle, so that it catches the egg	3
Modifying the properties of the egg	Freezing the egg before dropping it	5
Using natural properties of the egg	Dropping the egg on its most robust axis	0
Using the properties of the environment	Dropping the egg while no gravitation	3

natural properties of the egg (3%). Examples of solutions belonging to these expansive paths include training an eagle to catch the egg during the fall or freezing the egg before dropping it.

The solutions proposed by the participants were subsequently used with other new participants to determine whether examples belonging to the fixation path (i.e., restrictive examples) and examples belonging to expansive paths (i.e., expansive examples) would have opposite effects on the ability to generate creative ideas.

STUDY 2: THE IMPACT OF DIFFERENT EXAMPLES ON CREATIVITY

In the second study, we explored the impact of restrictive and expansive examples on creativity in the egg task. We hypothesized that the introduction of an example during the egg task would lead participants to propose more original solutions when the example provided was expansive (i.e., outside of the fixation effect). In contrast, when the example provided was restrictive (i.e., within the fixation effect), we hypothesized that the originality of the solutions given would be reduced. To measure the effect of those examples on creativity, we looked at the two criteria given by Guilford (1950) to assess creativity: the fluidity (the capacity to generate many solutions, i.e., the number of solutions) and the originality. As is common in divergent thinking literature, we computed an objective measurement of the originality of the solutions by taking the frequency of responses given across all the participants in each group. The originality of a solution was defined as the statistical infrequency of that particular solution. For example, if fewer than 5% of participants reported a particular response, this answer was considered to be of high originality.

On the contrary, if more than 95% of participants reported a particular response, this response was considered to be of low originality. A mean originality score was calculated for each participant, which could range from 0 to 1 (0 represented lower originality and 1 represented higher originality).

PARTICIPANTS

One hundred and thirty-one undergraduate students from Paris Descartes University were recruited for this study. The subjects were between 17 and 35 years old (mean age: 19.1 years). Participants were tested simultaneously in a large room. All participants were naïve regarding the experimental aims and had not participated in Experiment 1.

PROCEDURE

Each participant was randomly assigned to one of three experimental conditions (i.e., control without example, restrictive example, and expansive example) and was given 10 min to generate as many original solutions to one of the following problems.

Control group ($n = 78$): Without example

You are a designer and you are asked to propose as many original solutions as possible to the following problem: Ensure that a hen's egg dropped from a height of 10 m does not break.

Group A ($n = 27$): Restrictive example

You are a designer and you are asked to propose as many original solutions as possible to the following problem: Ensure that a hen's egg dropped from a height of 10 m does not break. One possible solution is to slow the fall with a parachute.

Group B ($n = 26$): Expansive example

You are a designer and you are asked to propose as many original solutions as possible to the following problem: Ensure that a hen's egg dropped from a height of 10 m does not break. One possible solution is to freeze the egg before dropping it.

RESULTS

To examine whether the fluidity and the originality varied according to the experimental conditions, we conducted one-factor analysis of variance (ANOVA) with the experimental groups (Control group, Group A, or Group B) as a between-subjects factor.

Regarding fluidity, that is, the number of solutions given per participant, the analysis revealed that the main effect of experimental group was close to reaching

significance ($F(2, 128) = 2.85, p = .06$), indicating that our experimental conditions tended to affect the number of solutions given by the participants. Further analyses revealed that the group exposed to the restrictive example proposed fewer solutions than the control group ($t(128) = 1.74, p < .05$, Cohen's $d = 0.39$) and the group exposed to the expansive one ($t(128) = 2.35, p < .05, d = 0.62$).

Regarding originality, the analysis showed a main effect of experimental group ($F(2,128) = 11.44, p < .005$), indicating that the presence of examples had a significant impact on the originality of the solutions (see Figure 2). Planned contrasts revealed that the solutions proposed by the group exposed to the restrictive example were less original than those given by the control group ($t(128) = 3.6, p < .005, d = 0.65$) and by the group exposed to expansive example ($t(128) = 4.7, p < .001, d = 1.04$). This suggests that exposure to a restrictive example reduced the originality of the solutions. On the contrary, our data showed that the group exposed to the expansive example proposed more original solutions than did the control group without an example ($t(128) = 2.13, p < .05, d = 0.59$).

Furthermore, we studied how the exposure to example impacted the distribution of solutions proposed by the participants regarding to the fixation effect identified in the first experiment. The analysis of our data shows that participants exposed to an expansive example managed to give more solutions outside of the fixation effect than those of the control group or those exposed to the restrictive example (See Table 2).

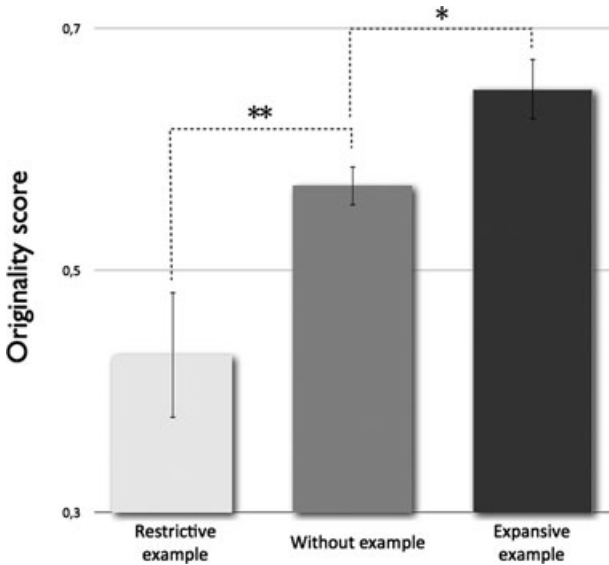


FIGURE 2. Originality score according to experimental group (with restrictive example; control without example; with expansive example). * $p < .05$; ** $p < .005$.

TABLE 2. Percentage of Answers Within and Outside Fixation, and Percentage of Answers on the Categories of the Chosen Examples

Percentage of answers	Control group, %	Restrictive example: the parachute, %	Expansive example: freezing an egg, %
Within Fixation	80	82	71
Outside Fixation	20	18	29
Slowing the fall	21	22	19
Modifying the properties of the egg	5	6	8
Other categories outside fixation	15	12	21

Indeed, solutions within the fixation effect represented 80% of the solutions for the control group, and this percentage dropped at 70% for the group with the expansive example (χ^2 significant at $p < .05$). We can, moreover, stress that the exposure to an expansive example did not recreate a new fixation effect around the proposed solution. Indeed, the expansive example “freeze the egg before dropping it” belonged to the category “modifying the properties of the egg.” The percentage of solutions in this specific category goes from 5% for the control group to 8% for the expansive example, but more solutions in other categories such as “acting before the fall” or “modifying the properties of the environment” are also given when participants are submitted to the expansive example.

DISCUSSION

The present experiments aimed at uncovering the impact of restrictive (i.e., belonging to the fixation effect) and expansive (i.e., outside of the fixation effect) examples on a creative task consisting of designing a way to drop a hen’s egg from a height of 10 m, so that it would not break. Three major findings emerged from this investigation: (1) there was confirmation that the ability of participants to generate creative ideas was blocked by solutions that spontaneously came to mind, leading to a fixation effect; (2) the introduction of an expansive example during the egg task led participants to propose more original solutions, whereas when the example provided was restrictive, both the number and the originality of solutions were reduced; and (3) the exposure to an expansive example stimulated participants, so that they gave more solutions outside the fixation effect.

Thus, our first study suggested that strong fixation effects do exist in creativity and that participants give solutions based on common knowledge that is spontaneously activated. This result is in line with studies in the domain of reasoning that have demonstrated the involvement of an intuitive/automatic system during problem solving (Allen & Thomas, 2011; Cassotti & Moutier, 2010; Cassotti et al., 2012; Kahneman & Frederick, 2007). In other words, when people have to solve a problem in a creative way, they are constrained by the automatic activation of common

knowledge, leading them to give usual solutions. Interestingly, when the activation of common knowledge was reinforced by the presentation of possible solutions belonging to the fixation path, participants proposed fewer, less original solutions. This finding is in agreement with several previous works, and confirms that recently activated knowledge can strongly constrain the ability to generate creative solutions (Jansson & Smith, 1991; Landau & Leynes, 2004; Landau et al., 2002).

One major result of the present investigation is that an expansive example produces a stimulation effect on originality. Indeed, providing an example not belonging to the fixation effect increases creative performance. The present result extends previous findings and reinforces the view that it is possible to enhance evocation processes using an external source of inspiration (Bonnardel & Marmèche, 2004; Bonnardel & Zenasni, 2007). However, although this result strongly suggests that a stimulating effect resulting from giving an example is possible (Fink et al., 2010), it can appear contradictory with regard to the vast number of studies reporting that examples block creativity (see the introduction section). Nevertheless, an explanation of this apparent discrepancy can be formulated thanks to the use of the C-K theory. Exposure to an expansive example unveils to the participants how expansions are possible in the C-Space, which is not the case for a restrictive example. Thus, an expansive example provides a set of alternatives in the C-Space, for example, using a living device or modifying the properties of the egg. However, this expansion could be limited to the C-Space in the sense that participants do not have access to methods of expanding their knowledge (for instance, via the internet). Moreover, to complete this explanation of stimulation effect offered by literature in design, work in cognitive psychology and neurosciences on inhibition (Houdé et al., 2000, 2001) also offers interesting perspectives: inhibition is a form of cognitive control that allows to resist automatism, intuitive reasoning, spontaneous activation of knowledge, and heuristics. The phenomenon of overcoming fixation effect can then be interpreted as a capability to inhibit spontaneously activated knowledge (leading to obvious solutions) to be able to expand and to explore original solutions. The exposure of example can then be a lever of action on activating inhibition processes in creativity.

A possible limitation of our study was that we computed an objective measurement of the originality of the solutions without taking into account the appropriateness of the solutions given by the participants. A creative idea is original, but an original idea is not always creative. Therefore, it would be interesting in future works to use an external rating procedure similar to the Consensual Assessment Technique proposed by Amabile (1996). Despite the marked interest in the study of the impact of external cues on creativity, there are many ambiguities in the field and little is known about the neuro-cognitive processes allowing for the control of routine strategies and the generation of new creative ones. Using both new techniques for modeling creative reasoning and recent advances in neuroimaging techniques, it would be possible in future work to directly observe the neural activity associated with the fixation effect and the neuro-cognitive impact of both restrictive and expansive examples.

In conclusion, the present results indicate that examples within or outside the fixation effect produce opposite effects on originality. These results provide experimental arguments enabling the reconciliation of studies that show either provocative or constraining effects of examples in creative settings.

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